SEEDLINE · CALCULATION NOTE

SeedLine sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: every fastener and the M16 press axle bolt weighed from the detailed model; mass fix SDL-DEC-001 F6 (3 mm drop and bearing plates, 20 x 2 mm uprights, aluminium marker arm, 1.5 mm disc) with its stress checks; push and mass figures re-run; requirements now in SDL-PRS-001; no requirement status changed
0.5	2026-10-02	Amish Chadha	Approved decisions carried into the design (SDL-DEC-001): 13 mm housing slot with printed side liners and a UV-stabilized door; mass, push force, cost and requirement figures re-run; no requirement status changed
0.4	2026-10-02	Amish Chadha	SDL-DDR-003 accepted and R17 cost accepted for the prototype (2026-10-02)
0.3	2026-10-01	Amish Chadha	Re-run on the constructable design (SDL-DDR-003); mass from the model's parts, hopper, depth geometry, push force, cost against value-engineering targets

0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (drive and spacing, metering, plates, mass, push force, slip, structure, work rate, depth, uniformity, cost) against every requirement

SeedLine sizing and first-principles checks

On paper the seeder works as a ground-driven single-row planter: the drive covers 22 to 589 mm of in-row spacing with the optional ratio kit, slip from the metering load is under 2 %, the parts are lightly stressed and the work rate is about 0.142 ha/h. Version 0.2 applies the recommendations Amish accepted on 2026-09-25 (SDL-DDR-002): a lighter 22 x 1.2 mm handle tube, the 12 and 18 T wheel sprockets moved to an optional ratio kit, a design-case walking speed of 2.9 km/h (the walking-speed rule for R1), and R8 accepted for field crops on ploughed land. Version 0.3 re-runs every figure on the constructable design of SDL-DDR-003 (2026-10-01), which adds the bearings, spacers, clips, brackets and fixings the concept left out and moves the opener forward of the metering housing; the mass is now taken part by part from the model. Version 0.5 carries the decisions of 2026-10-02 into the figures: the housing slot is 13 mm with two printed side liners, and the door is UV-stabilized polycarbonate. Version 0.6 follows the kit 1.9 detailed model, in which every bolt, nut, washer, pin and insert is modelled: weighing them from the model added about 0.6 kg to the seeder and took it to 15.5 kg with the marker kit, so under Amish's standing instruction of 2026-10-23 the drop plates and bearing plate are now 3 mm, the hopper uprights 20 x 2 mm flat bar and the marker arm aluminium (SDL-DEC-001 F6), and it is 14.8 kg again. The requirements are now in SDL-PRS-001, where each R number below is the "Was" of a PR. Eleven of the eighteen requirements are met on paper, R11 with a small margin (14.8 kg with the marker kit against 15 kg), and two cannot be verified at TRL 3. Three are **at risk**: R1 (cell speed 0.299 m/s at 2.9 km/h against an assumed 0.30 m/s limit, no margin), R2 (spacing CV about 31 % with assumed miss, double and scatter rates) and R10 (132 N horizontal push in the design case, but 179 N at best on a heavy, loose seedbed). None is not met. The two cost requirements are value-engineering targets (STANDARDS section 18): the prototype is USD 133.90 under its USD 400 target (R16), and the base seeder is USD 24.10 over its about USD 200 target (R17).

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the key dimensions, sprocket geometry, cell sizes, hopper volume and handle geometry from the parametric model `cad/src/model.py`, and the prices from `bom/bom.csv`, so the model, the drawing SDL-DWG-001, the BOM and this note agree. All values are first-principles estimates; nothing here is measured.

1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless marked as decided.

Input	Value	Basis
Architecture	Vertical cell plate, #35 chain, front drive wheel, rear press wheel, bolted frame, marker arm; 15 T wheel sprocket on the base seeder, 12 and 18 T in an optional ratio kit; 22 x 1.2 mm handle tube	Decided, SDL-DDR-001 items 1, 3, 4, 7 and 8; SDL-DDR-002 items 3 and 4
Constructable design	Drive axle turning in two flange bearings with the wheel fixed to it; plate shaft held from the chain side only; opener cross member ahead of the housing; spring tensioner; parts as in `cad/src/model.py`	SDL-DDR-003 (accepted by Amish, 2026-10-02)
Design case	Maize at 250 mm on 0.75 m rows, 50 mm deep, 2.9 km/h (0.81 m/s), the walking-speed rule for R1	SDL-REQ-001 v0.4; SDL-DDR-002 item 5 (was 3 km/h)
Drive wheel	300 mm over the lug tips, 45 mm wide, 10 mm lugs; travel 942.5 mm per turn before slip	Model parameter
Typical slip for spacing tables	5 %	Allowance; the calculated skid (section 6) is lower
Cell fill limit	0.30 m/s cell speed	Assumed working limit for a gravity-filled cell plate; to be checked by bench test at TRL 4
Cell size	Length 1.15 L + 0.5 mm, depth 1.05 W + 0.3 mm, plate thickness the larger of 6 mm and 1.2 T (L, W, T: seed length, width, thickness)	Rule used by `make_plate` in the model
Seed dimensions	Maize 12 x 8 x 5 mm, sorghum 4.5 x 4 x 3, groundnut kernel 15 x 9 x 8, common bean 13 x 8 x 6, spinach 3.5 x 3 x 2.5, pelleted carrot 3.3 mm, raw carrot 3 x 1.2 x 0.6	Typical values; varieties differ
Print	±0.2 mm on a 0.4 mm nozzle, tolerance limited to 10 % of the smallest cell dimension; 1.6 mm of skins, 1.2 mm walls, 20 % infill (25 % before SDL-DEC-001 F8); 6 mm³/s average rate; PETG 1.27 g/cm³ at $25/kg	Common desktop FDM printer
Meter torque	1.0 N·m at the plate shaft (design value)	First-principles estimate is 0.20 N·m (section 3); the design value is five times higher
Soil	Cone index 300, 200 and 100 kPa for good, design and heavy (loose, cloddy) seedbeds; opener specific resistance 20, 30 and 40 kPa; covering chains and press 10, 15 and 20 N	Assumed; no soil data from a partner region yet
Rolling resistance	Brixius motion resistance, MR/W = 1/Bn + 0.04, with Bn = CI b d / W / (1 + 3 b/d) for a rigid wheel	Brixius (1987), an empirical model fitted to larger tires; used here outside its calibrated range
Opener draft	Specific resistance x 2.5 x 18 mm opener width x depth (12 mm shank between two 3 mm boot plates)	Narrow-tool failure zone taken as 2.5 tool widths
Field work	50 m rows, 15 s per turn, 90 s per refill, 85 % working time	Assumed for a small plot and one operator
Structure	S235 class steel, yield 235 MPa; dynamic factor 3 for drops and stones; 100 N side load at the grip	Assumed

Input	Value	Basis
Uniformity model	5 % misses, 4 % doubles (second seed 0 to 30 mm behind), 15 mm drop scatter, slip 5 ± 1.5 %	The miss and double rates are the TRL 2 estimates; not calculable from first principles

2. Drive, spacing and range (R3, R4)

The spacing is $s = \pi D / (n\, i) / (1 - \sigma)$, with $\pi D = 942.5$ mm, n plate cells, ratio i = wheel teeth / 15 and travel reduction σ. The three wheel sprockets give i = 0.8, 1.0 and 1.2 (pitch diameters 36.80, 45.81 and 54.85 mm; plate sprocket 15 T). The base seeder carries the 15 T wheel sprocket (i = 1.0); the 12 and 18 T sprockets are an optional ratio kit (SDL-DDR-002 item 4).

Table 2. Example plates. Nominal spacing, and with the 5 % slip allowance.

Crop	Target (mm)	Cells	Ratio	Nominal (mm)	With 5 % slip (mm)
Maize (design case)	250	4	1.0	235.6	248.0
Sorghum, groundnut	150	6	1.0	157.1	165.3
Common bean	100	9	1.0	104.7	110.2
Spinach	50	16	1.2	49.1	51.7
Carrot, lettuce (pelleted)	25	30	1.2	26.2	27.6
Skip-cell plate	400	2	1.2	392.7	413.4
Skip-cell plate	450	2	1.0	471.2	496.0
Skip-cell plate	600	2	0.8	589.0	620.1
Shortest		36	1.2	21.8	23.0

With skip-cell plates (one or two cells; decided, SDL-DDR-001 item 5) and the ratio kit the nominal range is **21.8 to 589 mm**, and 1,178 mm with a single cell, so **R3 (25 to 400 mm) is met**. The settings are discrete: choosing the best plate and ratio for every target from 25 to 400 mm, the worst error is 11.1 % (a 372 mm target falls between 331 mm and 413 mm). The spacing card should list the nearest setting for each crop. The base seeder alone (15 T only) covers 26.2 to 471 mm nominal, but its settings are coarser at long spacings: the worst error is 17.3 % (a 400 mm target falls between 331 mm and 496 mm). The ratio kit is needed for the 25 mm pelleted-seed plate at 1.2 and for close settings above about 300 mm.

Chain. The sprocket centers are 286.2 mm apart. A #35 chain needs 73.61, 75.10 and 76.61 pitches for the 12, 15 and 18 T wheel sprockets, so the chain is set to 74, 76 or 77 links (the 77-link length needs one offset link). The largest slack left over is 8.6 mm (15 T, 76 links), which the spring tensioner takes up by deflecting the slack (lower) strand by about 35 mm. The tensioner in `bom/bom.csv` has about 45 mm of take-up, and the chain guard encloses its full travel.

3. Metering speed and cell fill (R1)

Amish accepted the walking-speed rule on 2026-09-25 (SDL-DDR-002 item 5): the operator walks at about 2.9 km/h with the 0.8 and 1.0 ratios and about 2.4 km/h with the 1.2 ratio. The design case now uses 2.9 km/h (it was 3 km/h in v0.1). At 2.9 km/h the wheel turns at 51.3 rpm. For the maize plate the seed centers sit on a 111.3 mm circle, so:

Table 3. Plate and cell speeds at 2.9 km/h.

Ratio	Plate speed (rpm)	Cell speed (m/s)	Walking speed for 0.30 m/s (km/h)
0.8	41.0	0.239	3.64
1.0	51.3	0.299	2.91
1.2	61.5	0.359	2.43

The design case runs at 0.299 m/s, at the assumed 0.30 m/s limit with no margin, and the limit itself is an assumption, so **R1 stays at risk**. A simple kinematic indicator (a still seed must fall the cell depth while the open cell passes it) gives 0.20 m/s for the maize cell; seeds in the pool are dragged along by the plate, so real plates fill at higher speeds, but the indicator confirms that cell speed, not the drive, sets the walking speed. The maize plate passes 3.4 cells per second; the 36-cell plate at 1.2 passes 36.9 cells per second at 2.9 km/h, which is demanding for the brush (the rule of 2.4 km/h at 1.2 lowers that further). Misses and multiples depend on seed shape, brush setting and cell speed and cannot be calculated on paper; they need the bench test that TRL 4 would bring.

Meter torque. Friction of the seed pool on the lower quarter of both plate faces (Janssen-type wall pressure under a 0.2 m seed head) is about 0.051 N·m, the brush about 0.048 N·m and the bearings and chain about 0.10 N·m, a total of **0.20 N·m**. The calculations below use 1.0 N·m, five times this estimate, to cover seed jamming at the brush.

A smaller plate would lower the cell speed in proportion to the cell circle. Under SDL-DDR-002 item 5 the plate size is revisited only with bench data, which is TRL 4 work and on hold.

4. Plates per crop (R5, R6)

Table 4. Cell sizes, print tolerance, mass and print time for each starter plate.

Plate	Cell length x depth (mm)	Plate thickness (mm)	Cells (maximum)	Tolerance / smallest cell dimension	Mass (g PETG)	Print time (h)
Maize	14.3 x 8.7	6	4 (21)	2.3 %	38.1	1.39
Sorghum	5.7 x 4.5	6	6 (47)	4.4 %	38.7	1.41
Groundnut kernel	17.8 x 9.8	10	6 (17)	2.1 %	51.4	1.87
Common bean	15.4 x 8.7	8	9 (20)	2.3 %	44.8	1.63
Spinach	4.5 x 3.5	6	16 (56)	5.8 %	39.4	1.44
Carrot, pelleted	4.3 x 3.8	6	30 (58)	5.3 %	40.3	1.47
Carrot, raw	3.9 x 1.6	6	30 (62)	**12.8 %, too fine**	40.1	1.46

Raw seed from 3.5 mm (spinach) to 15 mm (groundnut) fits the tolerance rule. Raw carrot seed needs a 1.6 mm deep cell, where ±0.2 mm is 12.8 % of the cell, so 1 to 2 mm seed must be sown pelleted (about 3.3 mm pellets), as decided (SDL-DDR-001 item 6). With that redefinition **R5 is met** on paper. Every starter plate prints in 2 h or less. The independent review of 2026-10-24 found that the print estimate counted only two sides of each cell's wall; counted correctly at 25 % infill the 10 mm groundnut plate took 2.05 h, a fail, so under the standing instruction the plate infill is now 20 % (SDL-DEC-001 F8), which the D-flat and jam stresses allow (section 7), and it takes 1.87 h. **R6 is met with a small margin** for thick plates, resting on an unsourced 6 mm³/s average rate (5 mm³/s would give about 2.2 h); the ±0.2 mm cell accuracy cannot be verified at TRL 3. The TRL 2 estimate of about 25 g and 1 h per plate was low; the model gives 38 to 51 g and 1.4 to 1.9 h.

5. Mass and center of mass (R11)

From v0.3 the steel and printed parts are weighed from the volumes of the constructable model (steel 7.85 g/cm³, PETG printed solid at 1.27 g/cm³, polycarbonate door 1.2 g/cm³, aluminium 2.70 g/cm³); bought parts keep catalogue-class estimates. From v0.6 the fasteners are weighed from the kit 1.9 model too, where each bolt, nut, washer, countersunk screw, set screw, clevis pin, split pin and insert is drawn to its ISO size, and the M16 press axle bolt replaces its 0.30 kg estimate.

Table 5. Mass by group.

Group	Mass (kg)	Notes
Wheels and axles (items 1, 11)	4.35	Drive wheel body 1.70 kg (bought estimate) and 18 lugs (model); drive axle, two 16 mm flange bearings (0.12 kg each) and spacers; press wheel 1.20 kg (bought estimate) with its M16 x 180 axle bolt, nut and washers (model) and spacers
Drive and metering (items 2 to 8)	2.42	Hubbed sprockets, chain and spring tensioner 0.75 kg; guard, hopper with lid, housing and the two plate liners from the model volumes; shaft, two flange bearings, collar, plate, brush, drop tube
Opener and covering (items 9, 10)	1.20	Shank and boot plates from the model; covering chains and bracket
Frame and handle (items 12, 13)	5.17	Frame 3.00 kg (two 700 mm rails, two cross members, four clips, four 3 mm drop plates, 3 mm bearing plate, four 20 x 2 mm uprights); handle 2.17 kg (two 974 mm side tubes of 22 x 1.2 round tube, grip, brace, sleeves, rubber grips)
Hardware (item 15)	1.08	Every fastener in the model except the press axle bolt and the marker kit's, 0.98 kg; paint and threadlocker 0.10 kg
Base seeder	**14.21**	
Row marker kit (item 14)	0.61	Mount, pivot spacers, 1.5 mm disc and fixings in steel, telescoping arm in aluminium tube 20 x 1.5 and 16 x 1.5 (model)
With marker kit	**14.82**	

Making the design constructable (SDL-DDR-003) added about 0.9 kg of bearings, spacers, clips, brackets, uprights and fixings and took off about 0.8 kg, and the 13 mm housing slot and its liners (2026-10-02) added 0.07 kg: 14.1 kg base and 14.9 kg with the marker kit in v0.5. The kit 1.9 model then showed that the fasteners weigh 0.98 kg, not

the 0.58 kg estimated for them with the paint, and that the M16 press axle bolt with its nut and washers weighs 0.38 kg, not 0.30 kg; with the marker kit's real pins and bolts the seeder came to 15.5 kg, over R11. The fix made under Amish's standing instruction (SDL-DEC-001 F6) takes off 0.72 kg: drop plates and bearing plate 3 mm instead of 4 mm (0.25 kg), hopper uprights 20 x 2 mm instead of 20 x 3 mm (0.19 kg), the marker arm in aluminium tube instead of steel (0.21 kg) and a 1.5 mm marker disc instead of 2 mm (0.06 kg); each is checked in section 7. **R11 is met, with 0.18 kg of margin** with the kit fitted (0.12 kg in v0.5); any further part needs a matching saving. In v0.1 the handle was 25 x 1.5 mm round tube, which put the seeder over 15 kg; the 22 x 1.2 mm tube Amish accepted on 2026-09-25 (SDL-DDR-002 item 3) saves 0.64 kg. The center of mass of the base seeder is 350 mm behind the drive axle and 243 mm above the ground; a full hopper adds 1.74 kg of maize.

6. Push force and slip (R10, R4)

The push is found from the planar statics of the seeder on its two wheels, with the operator's force at the grip (950 mm high, 647 mm behind the press wheel) at an angle φ below the horizontal. Rolling resistance acts at ground level and the opener draft 25 mm below it. For each case the script finds the smallest horizontal push that balances the resistance with both wheels on the ground.

Table 6. Push force (horizontal component) with a full hopper, 157 N total weight.

Seedbed	Along the handle (50°)	Lowest push, at angle	Angles that balance	Rolling / opener / chains (N)	Wheel loads, drive / press (N)
Good	81 N	72 N at 16°	11 to 50°	26 / 45 / 10	91 / 162
Design	**132 N**	110 N at 29°	22 to 50°	50 / 68 / 15	108 / 206
Heavy, loose	no balance	**179 N at 36°**	30 to 48°	69 / 90 / 20	216 / 70 (at 36°)

Three findings follow. First, a purely horizontal push has no balance in any case: at grip height it pitches the seeder forward and lifts the press wheel, so the TRL 2 comparison of a horizontal push against a push along the handle does not apply. Second, in the design case the horizontal push is 132 N when the operator pushes along the handle (the force along the handle is 206 N, 158 N of it downward; the independent review of 2026-10-24 found the earlier label "along the handle" misleading), inside the 150 N target of PR-010 with a 12 % margin that rests on assumed soil values; 68 N of it is opener draft from an assumed 30 kPa specific resistance. Third, on a heavy, loose seedbed a push along the handle drives the press wheel into the soil faster than it moves the seeder (no balance), and the best push is 179 N horizontal at 36°. **R10 is at risk.** The equivalent rolling coefficients are 0.102, 0.159 and 0.241, inside the 0.2 to 0.3 range assumed at TRL 2 only for the heavy case. Whether one operator can sustain 132 N at 0.81 m/s (about 106 W of work at the handle) for a planting day is not calculated here; it needs an ISO 11228-2 assessment (SDL-STM-001 row S9).

Slip. The drive wheel must supply the meter torque through the chain: 6.7 N at the rim at a 1.0 ratio and 8.0 N at 1.2. Against 91 to 210 N of wheel load, the Brixius traction relation gives a skid of **1.2 to 1.7 %**, well inside the 8 % of R4. The larger uncertainty is the rolling radius: if the lugs sink fully, the wheel rolls on its rim and travels up to 6.7 % less per turn, which makes the spacing shorter, not longer. The two effects partly cancel. **R4 is met on paper**; the spacing card should be calibrated by counting wheel turns over a measured 10 m on the user's own seedbed.

7. Structure

Table 7. Stress checks.

Part	Load case	Result
Side rails, 25 x 25 x 1.5 tube	Heavy case wheel and push loads, factor 3, plus opener moment	70 MPa, factor 3.3 on yield
Handle side tube, 22 x 1.2 round (SDL-DDR-002)	100 N side load at the grip	126 MPa, factor 1.9 (was 80 MPa, factor 2.9, with 25 x 1.5 tube)
Press axle, 16 mm	Heavy case load, factor 3	19 MPa
#35 chain	1.2 N·m on the 15 T plate sprocket	52 N against about 7.8 kN minimum tensile strength (typical catalog value)
Plate hub D-flat	1.2 N·m in PETG	5.7 MPa bearing stress
Rear drop plate, 3 mm (F6)	Pressing the handle down to lift the front at a row end, full hopper, factor 3; 50 mm wide plate bent in its own plane	18.7 N·m per plate, 45 MPa, factor 5.2
Hopper upright, 20 x 2 flat bar (F6)	Full hopper on four uprights, factor 3, each as a free-standing 302 mm strut	Euler load 72 N against 16 N, factor 5
Marker arm, aluminium 20 x 1.5 (F6)	20 N side drag at the disc, 650 mm lever	35 MPa, factor 6.9 on 6061-T6 (about 240 MPa)
Marker arm at full 900 mm reach (review)	20 N side drag, factor 3	128 MPa, factor 1.9 on 6061-T6; 6063-T6 (about 170 MPa) would give only 1.3, so the BOM names 6061-T6
Handle side tube, combined (review)	221 N heavy-seedbed push at 36° plus the 100 N side load	143 MPa, factor 1.6 on yield
Hopper upright sideways (review)	Three times the full hopper's weight sideways	Free-standing it would reach about 300 MPa, but the hopper's neck bears in the housing collar after 1 mm of sway: 10 MPa, the collar carries the rest
Jammed plate (review)	Wheel drives until it skids: about 11.4 N·m at the plate shaft	Chain 496 N against 7.8 kN; the plate's D-flat reaches about 54 MPa against about 50 MPa for PETG, so the printed plate's bore strips first and the steel drive is unharmed
Chain wear (review)	Chain at 1 % elongation	Needs 48 mm of idler travel against 45 mm of take-up: shorten by a link pair or replace the chain at about 0.8 % (build plan)

The frame and drive are lightly stressed. The independent review of 2026-10-24 (SDL-STM-001) added the last five rows. The lighter handle is the most highly stressed steel part, with a factor of 1.6 on yield under the side load combined with the heavy-seedbed push and no dynamic factor; fatigue of the telescoping joint is a TRL 4 question. Durability (R15) depends on wear of the plates, brush, chain and opener, which cannot be calculated at TRL 3.

8. Work rate and hopper (R9, R12)

The theoretical capacity on 0.75 m rows at the 2.9 km/h design walking speed is 0.217 ha/h. With 50 m rows, 15 s turns, 90 s refills and 85 % working time the field efficiency is 65 %, so the field capacity is **0.142 ha/h (7.1 h/ha)**, which meets R9 and compares with about 56 h/ha for hoe planting (SDL-PRB-001). The walking-speed rule costs about 0.003 ha/h against the v0.1 figure at 3 km/h (0.145 ha/h). On 0.3 m vegetable rows the capacity is 0.057 ha/h.

The hopper cavity in the model is **2.42 L** (R12 met), including the 12 mm straight neck that now drops into the housing collar. It holds 1.74 kg of maize, about 5,805 seeds, enough for 1,451 m of row or 0.109 ha per fill. Maize at 0.75 x 0.25 m needs 53,333 seeds per hectare.

9. Depth control (R8)

The opener shank is clamped to the opener cross member by two bolts into two of eight tapped holes at 10 mm pitch, which gives 10 to 60 mm in 10 mm steps. The front of the shank, the opener's leading edge, is 181 mm behind the drive axle (182 mm in the concept), so the frame, riding on its two wheels, raises the opener by 0.69 of any rise under the drive wheel and 0.31 of any rise under the press wheel. Holding ±10 mm therefore needs the wheel path to stay within about ±15 mm under the drive wheel or ±32 mm under the press wheel. A 50 mm clod under the drive wheel lifts the opener by 34 mm. Amish accepted the recommendation on 2026-09-25 (SDL-DDR-002 item 6): the rigid opener is accepted for field crops on ploughed land, and R8 is restated for a ploughed field-crop seedbed, where the ±15 mm wheel-path condition is expected to hold. On that basis **R8 is met on paper**; on rough, hand-tilled seedbeds depth will vary more, and a spring-loaded opener or depth-gauge shoe is to be reviewed with partner data.

10. Spacing uniformity (R2)

A Monte Carlo run of 20,000 cells for the maize plate, with the assumed rates in Table 1, gives a coefficient of variation of **30.7 %** for all plant-to-plant spacings, against a target of 30 % or less. The precision of the single spacings (those between 0.5 and 1.5 times the mean) is 8.9 %, with a miss index of 4.8 % and a multiple index of 3.9 %. The all-spacings CV is driven almost entirely by the misses and doubles, not by drop scatter or slip. **R2 is at risk**; it hinges on miss and double rates that only a bench test can give.

11. Cost (R16, R17)

All 16 BOM lines are priced; costs are estimates, not quotes. Both cost requirements are value-engineering targets, not limits (STANDARDS section 18; Amish, 2026-10-01).

- **R16, prototype.** Value-engineering target: USD 400 (`budget_usd`). Estimated cost of the constructable design: USD 266.10 for the base seeder with the marker kit (USD 24.00) and the optional ratio kit (USD 18.00), **USD 133.90 under the target**.
- **R17, base seeder.** Value-engineering target: about USD 200. Estimated cost of the constructable design: USD 224.10 for items 1 to 13 and 15, **USD 24.10 over the target**. The concept was USD 197.50; the parts that make it buildable add USD 22.00, and the decisions of 2026-10-02 add USD 4.60 more: the drive axle's two flange bearings and the lug strip (line 1, +6.00), hubbed sprockets and a bought spring tensioner (line 2, +5.00), the clear door, collar and knob (line 5, +4.00), then the decisions of 2026-10-02 (line 5, +4.60: the two printed liners about USD 1.60 of filament and the UV-stabilized door grade about USD 3.00 more than plain polycarbonate), more fasteners and the 36 lug screws (line 15, +3.00), the frame's clips, uprights and opener cross member (line

12, +2.00) and smaller items, less the opener clamp plate (line 9, −2.00). The main cost drivers and the savings worth trying are listed in the design decisions register (SDL-DEC-001, Value engineering).

12. Results against requirements

Table 8. Every requirement in SDL-PRS-001 v0.8 (by its old R number, the PR's "Was"), at-risk items first (none is not met).

ID	Requirement	Value (this note)	Target	Status
R1	Single-seed placement	Cell speed 0.299 m/s at the 2.9 km/h walking-speed rule against an assumed 0.30 m/s limit, no margin	Misses and multiples 5 % or less each	**At risk**
R2	Spacing uniformity	CV 31 % of all spacings; 9 % for singles (assumed rates)	CV 30 % or less	**At risk**
R10	Push effort	132 N horizontal pushing along the handle (design case; 206 N along it); 179 N horizontal at best on a heavy, loose seedbed	150 N or less	**At risk**
R7	Crop change	Side door and hand knob in the model; one 13 mm slot with a liner pair for the 6, 8 and 10 mm plates	2 min or less, no tools	Not verifiable at TRL 3
R15	Durability	Stresses low (section 7); wear life unknown	5 seasons; plates 1 season	Not verifiable at TRL 3
R11	Mass and handling	14.2 kg base; 14.8 kg with the marker kit (fasteners from the model, fix F6)	15 kg or less	Met, 0.18 kg margin
R17	Replication cost	USD 224.10 base seeder	About USD 200 (value-engineering target)	USD 24.10 over the target; USD 219.50 accepted for the prototype on 2026-10-02, USD 4.60 added by the liners and the UV-stabilized door
R8	Sowing depth	10 to 60 mm in 10 mm steps; ±10 mm for wheel-path bumps within about ±15 mm	±10 mm on a ploughed field-crop seedbed (restated)	Met on paper
R3	Spacing range	22 to 589 mm nominal with the ratio kit, every 25 to 400 mm target within 11.1 %; base seeder 26 to 471 mm, within 17.3 %	25 to 400 mm	Met (with the ratio kit)
R4	Spacing follows travel	Skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %	Slip 8 % or less	Met
R5	Seed size range	3.5 to 15 mm raw seed; 1 to 2 mm seed as pellets of about 3.3 mm	1.5 to 15 mm, seed under 2 mm pelleted	Met
R6	Printable plates	1.39 h and 38 g (maize); up to 1.87 h (groundnut); accuracy not verifiable	2 h or less	Met, small margin for thick plates
R9	Work rate	0.142 ha/h (7.1 h/ha)	0.1 ha/h or more	Met
R12	Hopper	2.42 L; 1,451 m of maize row per fill	2 L or more	Met
R13	Row-spacing kit	Marker reach 200 to 900 mm, set at 750 mm in the model	200 to 900 mm, ±25 mm	Met (design review)

ID	Requirement	Value (this note)	Target	Status
R14	Guarding	Printed shroud with an outboard face plate over both sprockets, the chain and the tensioner's full travel	Nip points covered	Met (design review)
R16	Prototype cost	USD 266.10	USD 400 value-engineering target	USD 133.90 under the target
R18	Local build	Bolted 25 mm square tube, 3 mm plate, angle and flat bar, #35 chain; saw, drill, tap and bolts	Common sections, drill and bolts	Met (design review)

Summary: 11 met, 3 at risk, 0 not met, 2 not verifiable at TRL 3, and the two cost requirements reported against their value-engineering targets (R16 under, R17 over). (v0.2: 13 met, counting both cost requirements as met; v0.1: 10 met, 4 at risk, 2 not met, 2 not verifiable.)

13. Corrections to the TRL 2 figures

Quantity	TRL 2 (SDL-PRC-001 v0.2)	TRL 3 (v0.1 of this note; see Table 14 for v0.2)
Spacing range	About 22 to 393 mm; R3 not met	21.8 to 589 mm with skip-cell plates; R3 met
Plate mass and print time	About 25 g, about 1 h	40 to 54 g, 1.5 to 2.0 h
Hopper	About 2.1 L, about 1,250 m per fill	2.40 L, 1,438 m per fill
Mass	13.1 kg base, 14.1 kg with marker	14.6 kg base, 15.6 kg with marker
Push force	About 100 N good, about 220 N heavy	83 N good, 136 N design, 178 N heavy at best
Slip	3 to 8 %	1.2 to 1.7 % skid, plus up to 6.7 % rolling radius uncertainty
Work rate	About 0.14 ha/h	0.145 ha/h
Parts cost	$199 base, $223 with marker	$214 base, $238 with marker
Meter torque	0.5 to 1.0 N·m	0.20 N·m estimated; 1.0 N·m used

Table 14. Changes in v0.2 from the recommendations accepted by Amish (SDL-DDR-002).

Quantity	v0.1	v0.2
Handle tube	25 x 1.5 mm, 2.99 kg handle	22 x 1.2 mm, 2.34 kg handle
Mass	14.6 kg base, 15.6 kg with marker (R11 not met)	13.9 kg base, 14.9 kg with marker (R11 met)
Base parts cost	$213.50 (R17 not met)	$197.50 (R17 met); ratio kit $16.00 optional
Design walking speed	3 km/h, cell speed 0.309 m/s	2.9 km/h, cell speed 0.299 m/s
Work rate	0.145 ha/h	0.142 ha/h
Push, design case	136 N along the handle	132 N along the handle
Push, heavy seedbed	178 N at 35°	177 N at 36°
Handle side tube	Factor 2.9 on yield	Factor 1.9 on yield
R8	At risk	Restated for ploughed field-crop seedbeds; met on paper

Table 15. Changes in v0.3 from the constructable design (SDL-DDR-003).

Quantity	v0.2	v0.3
Mass	13.9 kg base, 14.9 kg with marker (0.06 kg margin)	14.0 kg base, 14.8 kg with marker (0.19 kg margin), weighed from the model
Hopper	2.40 L, 1,438 m per fill	2.42 L (straight neck added), 1,451 m per fill
Opener leading edge	182 mm behind the drive axle	181 mm; depth figures unchanged
Push, design case	132 N along the handle	132 N along the handle
Push, heavy seedbed	177 N at 36°	178 N at 37°
Base parts cost	USD 197.50 (reported as met)	USD 219.50, USD 19.50 over the about USD 200 value-engineering target
Prototype parts cost	USD 237.50	USD 261.50, USD 138.50 under the USD 400 value-engineering target
Slack take-up	Spring idler	Bought spring tensioner, 45 mm take-up, inside the guard

> **Safety:** These figures are paper estimates. The chain drive, opener point and wheel lugs remain hazards whatever the numbers say; the guard must be fitted in use and plate changes done with the wheel held still (SDL-PRC-001, Safety). Treated seed is toxic; follow the seed label.

Table 16. Changes in v0.5 from the decisions of 2026-10-02 (SDL-DEC-001).

Quantity	v0.4	v0.5
Housing slot	9 mm, for the 6 mm plate only	13 mm, with printed side liners for the 6, 8 and 10 mm plates
Mass	14.0 kg base, 14.8 kg with marker (0.19 kg margin)	14.1 kg base, 14.9 kg with marker (0.12 kg margin)
Push, design case	109 N at 31°	110 N at 29° (132 N along the handle, unchanged)
Base parts cost	USD 219.50, USD 19.50 over the about USD 200 target	USD 224.10, USD 24.10 over the target (liners USD 1.60, UV-stabilized door grade USD 3.00)
Prototype parts cost	USD 261.50, USD 138.50 under the USD 400 target	USD 266.10, USD 133.90 under the target
Requirement statuses		None changed

Table 17. Changes in v0.6 from the kit 1.9 detailed model and the mass fix (SDL-DEC-001 F6).

Quantity	v0.5	v0.6
Fasteners	0.58 kg estimate (about 50 M6 and M8 sets, 36 lug screws, inserts, paint)	0.98 kg weighed from the model, plus 0.10 kg paint and threadlocker
Press axle bolt	0.30 kg estimate	0.38 kg from the model (M16 x 180 with nut and washers)
Drop plates, bearing plate	4 mm	3 mm (F6)
Hopper uprights	20 x 3 mm flat bar	20 x 2 mm (F6)
Marker arm, disc	Steel tube 20 x 1.2 and 16 x 1.2; 2 mm disc	Aluminium tube 20 x 1.5 and 16 x 1.5; 1.5 mm disc (F6)
Mass	14.1 kg base, 14.9 kg with marker (0.12 kg margin)	14.2 kg base, 14.8 kg with marker (0.18 kg margin); 15.5 kg with the marker before F6
Push, design case	132 N along the handle	132 N horizontal pushing along the handle (label corrected)
Push, heavy seedbed	178 N at 36°	179 N at 36°
Requirement statuses		None changed